

API Documentation

REST API reference for SpaceCDF backend

SpaceCDF -- AI-Assisted Concurrent Design Facility

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DRAFT

SpaceCDF API Documentation

Base URL: <http://localhost:8000/api>

Total Endpoints: 105+

Format: JSON (request and response)

Core Design

POST /design/quick-design

Run the full design loop (20 agents, convergence in seconds).

Request:

```
{
  "requirements": {
    "name": "My Mission",
    "mission_type": "earth_observation",
    "spacecraft_class": "nano",
    "orbit": { "orbit_type": "sso", "altitude_km": 500, "inclination_deg": 97.4 },
    "payloads": [{ "name": "Imager", "mass_kg": 1.5, "power_w": 8, "data_rate_mbps": 100 }],
    "design_lifetime_years": 3
  }
}
```

Response: { converged, iterations, total_time_s, parameters: {id: {value, unit, domain}}, budgets, warnings, conflicts }

POST /studies/

Create a study (persists requirements + mission need for later reference).

GET /studies/{id}

Retrieve study state.

Lifecycle Services

POST /lifecycle/orbit-trade

Compute orbit trade study with scored candidates.

Request: { target_gsd_m, target_revisit_days, target_latitude_band, aperture_m, max_cost_meur, min_lifetime_years, mission_type }

Response: { candidates: [{ name, altitude_km, inclination_deg, achievable_gsd_m, revisit_days, natural_lifetime_years, total_score, rank }] }

POST /lifecycle/class-advisor

Recommend spacecraft class from performance/programmatic targets.

POST /lifecycle/mission-trade

Evaluate space vs non-space alternatives (filtered by mission_type, includes constellation option).

GET /lifecycle/spectrum/bands

Available frequency bands filtered by mission type and license type.

Query params: mission_type, license_type (amateur/experimental/commercial), data_rate_mbps

POST /lifecycle/spectrum/itu-api-template

Generate ITU Advance Publication Information filing template.

POST /lifecycle/spectrum/iaru-template

Generate IARU amateur coordination request template.

POST /lifecycle/regulatory/rssa

Generate Canadian RSSSA filing template.

POST /lifecycle/regulatory/export-assessment

Generate export control classification assessment.

POST /lifecycle/regulatory/copuos-registration

Generate UN Registration Convention Article IV template.

POST /lifecycle/regulatory/eol-report

Generate end-of-life analysis report.

GET /lifecycle/parametric-data

Return all parametric model data (mass fractions, cost fractions, power duty cycles, SA power tables) with source citations.

POST /lifecycle/duty-cycles

Estimate power duty cycles for a mission configuration.

Request: { spacecraft_class, mission_type, comms_band, eclipse_fraction }

GET /lifecycle/consistency/{study_id}

Run full design consistency check. Returns health score, issues by category.

POST /lifecycle/trade-study

Run a tabular trade study with criteria, weightings, and options.

Request: { name, criteria: [{id, name, weight, direction}], options: [{id, name, scores: {criterion_id: value}}] }

GET /lifecycle/trade-templates

List pre-built trade study templates (orbit, component, ground, architecture).

POST /lifecycle/constellation/design

Design Walker delta constellation for coverage targets.

GET /lifecycle/beyond-leo/orbits

List beyond-LEO orbit options (MEO, GEO, HEO, Lunar, interplanetary) with environment data.

POST /lifecycle/beyond-leo/transfer

Compute transfer orbit ΔV from LEO to target orbit.

POST /lifecycle/beyond-leo/dsn-link

Compute deep-space link budget for DSN communication.

GET /lifecycle/requirements/generate/{study_id}

Generate SMART requirements from study objectives and functions.

POST /lifecycle/requirements/validate

Validate a requirement against SMART criteria.

POST /lifecycle/requirements/check-compliance

Check non-compliance and get resolution options.

Engineering Services

GET /engineering/equipment/{domain}/search

Search KB for equipment compatible with current design.

Domains: power, aocs, link, propulsion, structure, data, thermal, integration

GET /engineering/equipment/needs/{study_id}

Determine which equipment categories are needed based on requirements.

POST /engineering/equipment/check-compatibility

Check RF compatibility between transponder and antenna.

POST /engineering/equipment/budget-impact

Compute live budget impact of current equipment selections.

GET /engineering/verification

Run requirement verification against current design state.

GET /engineering/cost

Get cost estimate for current design.

POST /engineering/analysis/sensitivity

Run parametric sensitivity sweep.

GET /engineering/analysis/eol-curves

Get end-of-life degradation curves.

POST /engineering/analysis/trade-study

Run equipment-level trade study.

POST /engineering/impact-preview

Preview downstream impact of parameter changes (dry run ? no execution).

ECSS Compliance

GET /ecss/phases

List available ECSS phases.

GET /ecss/compliance/{phase_id}

Get expected DRDs and SpaceCDF coverage for a phase.

GET /ecss/margins/{study_id}

Check all budget margins against ECSS phase-appropriate policy.

GET /ecss/dids

List available DID (Document Item Description) types.

POST /ecss/dids/{did_type}/generate

Generate an ECSS DID document. Types: mrd, ts, ird, semp, rmp, conops, test_plan.

Optimizer

GET /optimize/config

Get available objectives and design variables (filtered by mission_type if provided).

Query params: mission_type, has_propulsion, pointing_accuracy_deg

POST /optimize/sessions/{session_id}

Start an optimization run (single-objective or Pareto NSGA-II).

GET /optimize/runs/{run_id}

Get optimization run status and results.

POST /optimize/sensitivity/{session_id}

Run Morris screening sensitivity analysis.

Sessions (Concurrent Design)

POST /sessions/

Create a new design session.

GET /sessions/{id}

Get session state.

GET /sessions/{id}/history

Get edit history for a session.

WebSocket /ws/session/{session_id}

Real-time collaboration. Message types: parameter_update, state_update, convergence_complete, participant_joined/left.

Positions

GET /positions/

List all engineering positions with roles and key questions.

GET /positions/{id}/guidance

Get computed guidance for a position based on current design.

POST /positions/answers

Save a position question answer.

GET /positions/answers

Get all saved position answers.

Knowledge Base

GET /kb/components/{category}

Get all components in a category.

GET /kb/categories

List all component categories.

Exports

POST /exports/docs/{format}

Generate design review documents (SRR, PDR, CDR).

POST /exports/mbse/{study_id}

Generate MBSE JSON export (ECSS-E-TM-10-25A style).

POST /exports/fsw/{study_id}

Generate flight software architecture (mode manager, FDIR, TC/TM).